

Course Code: CSE220	Course Title: Internet of Things Type of Course: Integrated	L- T-P- C	1	0	4	3
Version No.	2.0					
Course Pre-requisites	1. Students should know basic python programming. 2. Students have basic knowledge basic electronic components such as sensors – temperature, motion, pressure, and actuators etc. 3. Students should have basic idea about Cloud and its uses.					
Anti-requisites	NIL					
Course Description	The Internet of Things (IoT) is an emerging paradigm combining heterogeneous devices at an unprecedented scale, thereby enabling individuals and organizations to gain greater value from networked connections among people, processes, data, and things. The Internet of Things (IoT) is a course of objects interacting with people, with information systems, and with other objects. The course will focus on creative thinking, IoT concepts & IoT technologies.					
Course Objective	The objective of the course is to familiarize the learners with the concepts of Internet of Things and attain SKILL DEVELOPMENT through EXPERIENTIAL LEARNING techniques					
Course Out Comes	On successful completion of the course the students shall be able to: Identify the application areas of IoT Understand building blocks of Internet of Things and characteristics Describe IoT Protocols Demonstrate use of IoT devices for simple application					
Course Content:						
Module 1	INTRODUCTION TO INTERNET OF THINGS	Assignment	Simulation/Data Analysis		18 Sessions	
Introduction, Definition & Characteristics of IOT, Physical Design of IoT- Things in IoT, IoT Protocols, Logical design of IoT- IoT functional blocks, IoT Communication Models, IoT Communication APIs, IoT Enabling Technologies- Wireless sensor networks, Cloud computing, Big data Analytics						
Module 2	IOT COMMUNICATION MODEL AND PROTOCOLS	Assignment	Numerical from E-Resources		18 Sessions	
Connectivity Protocols: 6LoWPAN, IEEE 802.15.4, Zigbee, Wireless HART, Z-Wave, ISA 100,NFC, RFID. Communication/Transport Protocols: Bluetooth. Data Protocols: Message Queue Telemetry Transport (MQTT), Constrained Application Protocol (CoAP), Advanced Message Queuing Protocol (AMQP), XMPP – Extensible Messaging and Presence Protocol						

Module 3	IOT COMMUNICATION MODEL AND PROTOCOLS	Term paper/Assignment	Simulation/Data Analysis	19 Sessions
Communication/Transport Protocols: Bluetooth. Data Protocols: Message Queue Telemetry Transport (MQTT), Constrained Application Protocol (CoAP), Advanced Message Queuing Protocol (AMQP), XMPP – Extensible Messaging and Presence Protocol. RFID: Introduction, Principle of RFID, Components of an RFID system.				
List of Laboratory Tasks 1 Installation of arduino IDE & Arduino program to implement scrolling LED, to glow even/odd LED 2 Arduino program to demonstrate usage of push button to control the LED 3 Arduino program to demonstrates traffic control system 4 Arduino program to demonstrates usage of servo motor with potentiometer. 5. Arduino program to Control an LED using Bluetooth. 6. Arduino program to implement RFID reader for security access. 7. Arduino Program to detect obstacle using IR sensor. 8. Arduino Program to detect motion using PIR sensor. 9. Installation of Raspberry pi software 10. Working basic commands on Raspberry pi & to demonstrate remote logging in raspberry pi 11. Raspberry pi program to implement blinking LED 12. Raspberry pi program to implement camera module for video 13. Raspberry pi program to obtain the temperature using DHT sensors 14. Using a Raspberry Pi with distance sensor (ultrasonic sensor HCSR04) 15. Raspberry pi program to implement Garage spot light				
Targeted Application & Tools that can be used: Interfacing of ARDUINO and Raspberry pi for developing smart CITIES Tools: Tinker cad Cooja simulator Contiki Thingspeak				
Text Book T1 Arshdeep Bagha, Vijay Madiseti, Internet of Things A hands on approach, First Edition, Universities Press, 2018 T2 Hakima Chaouchi, The internet of Things Connecting Objects to web Wiley 2017				

References

R1 Vinit Kumar Gunjan, MohdDilshad Ansari, Mohammed Usman, ThiDieuLinh Nguyen Internet of

Things Technology, Communications and Computing Springer January 2023

R2 Dr. Hassan Internet of Things A to Z: Technologies and Applications IEEE Press 2018

E-Resources

NPTEL course –

a) https://onlinecourses.nptel.ac.in/noc22_cs53/preview

b) <https://www.udemy.com/course/complete-guide-to-build-iot-things-from-scratch-to-market/>

c) <https://puniversity.informaticsglobal.com:2229/login.aspx>

Topics relevant to “SKILL DEVELOPMENT”: Case studies of water supply projects – Design criteria through group discussion. Interpolation of sensors through group presentation for Skill Development through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.